Task Discovery Framework
An Agent-Initiated Discovery Approach

IETF Side Meeting — AI Ecosystem & Discovery

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https://datatracker.ietf.org/doc/draft-mapmw-task-discovery/

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Motivation for this Architecture

- **Current State: Discovery is "Agent-Centric“ and broad-spaced**
  - Agents are long-lived entities with static profiles such as agent cards.
  - There is a rapidly growing interest in using agentic services to handle an increasing **variety** and **volume** of tasks across a wide range of domains.
  - Task owners search registries for agents based on IDs, static profiles, or capabilities.
  - Example: A2A, ARDP, DNS-AID, and AGNTCY.

- **The Problem:**
  - **Scalability & Freshness:** Storing and replicating profiles for billions of agents globally is costly, and profiles can become stale.
  - **Friction**: Task owners often don't know the exact technical and agentic skills required for a given task as this requires expertise in the domain of the task.
  - **Task Magnets & Cold-Start:** Visibility favors established agents; newcomers struggle to get work and build reputation.
  - **Complex Task Handling:** Owners are burdened with decomposing complex tasks into subtasks without knowing which agents exist.

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Shifting the Paradigm: Task Discovery

- **Concept:**
  - Agent-initiated engagement flips the discovery model. Let the tasks be discoverable, and let the agents come to the tasks.

- **Key Concept:**
  - Task Owners post "Task Cards" to a Task-Posting Platform (e.g., a billboard).

- **New Role for Agents:**
  - Agents actively scan these billboards, find tasks that match their skills, and apply to complete them.

- **Vision:**
  - An open ecosystem where task owners describe what they need, and agents figure out how to complete posted tasks relying on their onboard intelligence.

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Task discovery framework in a nutshel

- **How it works:**
  - Instead of "searching for an agent," the owner "posts a task” to some sort of a platform. The task is represented as a structured “Task Card”.
  - Agents can then discover tasks posted to this platform, select those suiting their skillset and capabilities, and potentially connect to task owners to ask for task assignment.
- **Why now?**
  - The explosion in interest for utility of agentic services within a broad and heterogeneous domain space of tasks
- **Benefits:**
  - Demand-Driven: Task records are ephemeral; they expire once the task is done (reduces storage).
  - Expertise Shift: Agents (who know their own capabilities best) evaluate if they can do the job. Also helps agents self-improve but acquiring the most needed set of skills given the task demographics
  - Dynamic Bidding: Enables competitive bidding and better load balancing for idle agents.

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The Task Card (The Core Data Model)

- **What is a Task Card?**
  - A structured metadata object describing the work (possibly standardized)
  - Just like agent cards, they are discoverable objects (*can be carried as part of the discovery mechanism presented in draft-am-layered-ai-discovery-framework)
- **Key Fields: Potential for standardization??**
  - **Task Metadata:** ID, timestamp, expiration, priority.
  - **Requirements:** Specific skills, geographic constraints, or hardware needs.
  - **SLAs/Constraints:** Deadline, budget, or security requirements.
  - **Communication:** How the agent should respond (e.g., a callback URL or public key).
  - **Handshake Info:** Public keys or callback URLs for the agent to respond.
- **Why it's powerful:**
  - Allows non-expert owners to post raw, unsplit tasks. Shifts the interpretation burden to the intelligent agents.

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Architectural Framework Layers

- **Task Owner Layer:** The entities that need work done.
- **Task Owner Access Layer:** Standardized APIs for owners to post/manage tasks.
- **Task-Posting Platform:** The "Billboard" or "Exchange" (e.g., specialized registries, DHTs, or social media-like platforms).
- **Agent Access Layer:** Standardized APIs for agents to discover, search, and apply for tasks.
- **Agentic Layer:** Autonomous agents scanning for tasks.
- **Communication Link (Optional):** A secure, direct channel for post-discovery negotiation (clarifications, contracts, progress updates) between owner and agent.

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|| T.O. 1 || <--> |        |                        |        | <--> || Agent 1||
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|+--------+|      |  T.O.  |      |   Task    |      | Agent  |      |+--------+|
|| T.O. 2 || <--> | Access | <--> |  Posting  | <--> | Access | <--> || Agent 2||
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* T.O: Task Owner
```

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Operational Workflow - sample

- **Publishing:** Owner posts Task Card to the platform.
- **Discovery:** Agent discovers the card via search or subscription. (APIs can be exposed by the task posting platform that allows agents to properly discover tasks)
- **Evaluation:** Agent autonomously checks its own skills against the Task Card requirements.
  - In some scenarios once an agent discovers a task that might be complex, the agent initiates an agent **coalition formation procedure** where it might need to discover other agents- potentially facilitated by **traditional agent discovery mechanisms**
- **Proposal/Bidding:** Agent submits a proposal (cost, time, credentials).
  - In some scenarios, depending on task card, no proposal or bidding might be needed
- **Selection & Bilateral Handshake:** Owner selects an agent; both establish a secure link for execution.
  - In case of a task being handled by a coalition, owners might request IDs of all agents in the coalition or might be okay with one representative agent

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Key Advantages of the Framework

- **Scalability & Efficiency:**
  - Storage is demand-driven. Active task cards expire, unlike persistent agent profiles. Infrastructure cost aligns with actual workload. Additionally, allows for hierarchical operation (dispatcher agents, domain based grouping…etc)
- **Reduced Owner Burden:**
  - Owners don't need to be experts. They post the problem; agents figure out fitness and propose a solution plan. Shifts diagnostic intelligence to the provider side.
- **Solves the Cold-Start Problem:**
  - New agents can proactively find and apply for tasks, building reputation through completed work instead of waiting to be chosen.
- **Natural Handling of Complex Tasks:**
  - Agents can discover a complex task ("build a bicycle") and dynamically form coalitions amongst themselves to handle all subtasks, hiding that complexity from the owner.

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Relationship with Existing Protocols

- **Complementary, not Replacement:**
  - ARDP/A2A: Can handle the secure control-plane and data-plane between agents after task discovery. ARDP can provide secure discovery, while A2A can handle agent-to-agent communication.
  - DNS-AID: although can be used for registering and discovering agents, they could find a role in registering task cards and making them discoverable.

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Discussion & Next Steps

- **Goals for the Draft:**
  - Define the standard schema for Task Cards.
  - Standardize the Bilateral Handshake interface between owner and agent.
  - Look into standardizing interactions
    - between task owners and task posting platforms
    - between agents and task posting platforms
- **Questions for the Room:**
  - Is there interest in a unified Task Discovery model?
  - Where does this work best live (which WG)?
  - Are there existing data models (like JSON Schema for AI) we should align with?

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